

DILEEPA D

Portfolio Profile & Content Blueprint

Computer Science Engineering • Data Science • AI/ML • Generative AI • Big Data

1. PORTFOLIO IDENTITY

Name: Dileepa D

Location: Bengaluru, Karnataka, India

Degree: B.E. Computer Science and Engineering

College: Don Bosco Institute of Technology

Expected Graduation: 2027

CGPA: 8.2 / 10

Target Career: Data Scientist, with strong alignment to Python/Backend, AI/ML and Generative AI roles

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2. PORTFOLIO HERO SECTION

Recommended headline: Computer Science Engineer | Data Science | AI/ML | Generative AI

Recommended introduction: Computer Science Engineering student with a strong interest in Data Science, Machine Learning, Artificial Intelligence, Generative AI, NLP, and Big Data. Experienced in Python-based data analysis, deep learning, computer vision, sentiment analysis, RAG and LangChain workflows, with hands-on exposure to Hadoop, Spark, Flask, TensorFlow, and modern data/AI tooling.

3. ABOUT ME

I am a Computer Science and Engineering student at Don Bosco Institute of Technology, focused on building practical solutions with Python, data, machine learning and AI. My project work spans real-time facial emotion recognition, Netflix data analysis, large-scale social media sentiment analysis, and Retrieval-Augmented Generation workflows. I enjoy moving from raw data and experimentation toward usable applications and interfaces.

4. TECHNICAL SKILLS

Category	Portfolio-ready skills
Programming	Python, SQL, HTML, CSS, JavaScript
Data Science	Pandas, NumPy, Matplotlib, Seaborn, EDA, data cleaning, preprocessing, visualization
Machine Learning	Scikit-learn, regression, classification, KNN, Decision Trees, Naive Bayes, K-Means, PCA, SVD
Deep Learning	TensorFlow, Keras, CNN, image classification
Computer Vision	OpenCV, DeepFace, facial emotion recognition
NLP	TF-IDF, Word2Vec, Gensim, sentiment analysis, text classification
Generative AI	RAG, LangChain, LLMs, embeddings, semantic search, prompt engineering
RAG Stack	Vector search, Sentence Transformers, document retrieval, context-aware generation
Big Data	Apache Hadoop, Apache Spark, distributed data processing
Backend / Deployment	Flask, Gradio
Databases	MySQL, PostgreSQL, MongoDB/BSON fundamentals
Tools	Git, GitHub, Jupyter Notebook, Google Colab, VS Code

5. FEATURED PROJECTS

Real-Time Face Emotion Recognition System

Technology: Python • TensorFlow • OpenCV • CNN • Deep Learning

- Developed a real-time facial emotion recognition system using live facial expressions.
- Used CNN/deep learning and OpenCV for image processing and emotion classification.
- Trained and evaluated an image-classification model and integrated it into an end-to-end application.

Netflix Movies & TV Shows Data Analysis

Technology: Python • Pandas • NumPy • Matplotlib • Seaborn

- Analyzed a dataset of 8,000+ Netflix titles to identify content trends and release patterns.
- Performed data cleaning, preprocessing and exploratory data analysis.
- Built a content-based recommendation system and communicated findings through visualizations.

Big Data Social Media Sentiment Analysis & User Behavior Mining

Technology: Python • Hadoop • Spark • NLP • TF-IDF • Naive Bayes • Pandas

- Analyzed large-scale social media data to study sentiment patterns and user behavior.
- Applied TF-IDF and Naive Bayes-based classification for sentiment analysis.
- Explored Hadoop and Apache Spark for distributed processing of large datasets.

RAG / LangChain AI Pipeline

Technology: Python • RAG • LangChain • LLMs • Embeddings • Sentence Transformers

- Studied and worked with Retrieval-Augmented Generation for context-aware responses from external documents.
- Worked with LangChain, embeddings, semantic search, vector-based retrieval and Sentence Transformers.
- This project area can be expanded into a flagship portfolio project with document upload, retrieval, citations and evaluation.

6. ADDITIONAL PROJECT / ANALYTICS EXPERIENCE

- CNN image classification workflow: dataset preparation, model compilation, training, testing, confusion matrix, classification report and deployment with Gradio.
- EPA Green Vehicle Guide analysis: data loading, encoding/error handling, filtering, cleaning, visualization and analysis.
- Business Sales Performance Analytics: exploratory analysis, cleaning, visualization and business insight generation.
- Customer Retention & Churn Analysis: customer behavior, churn patterns, retention analysis and visualization.
- Marketing Funnel & Conversion Performance Analysis: funnel stages, conversion metrics, cleaning, EDA and business insights.

7. EDUCATION

Don Bosco Institute of Technology

Bachelor of Engineering (B.E.) — Computer Science and Engineering

Aug 2023 – Jun 2027 (Expected) • CGPA: 8.2 / 10

8. CERTIFICATIONS & COURSEWORK

- Introduction to Generative AI — SkillUp
- Software Engineering Job Simulation — JPMorgan Chase, via Forage
- BASH Training and RDBMS PostgreSQL Training — Edupyramids
- Software Project Management — NPTEL, IIT Kharagpur
- Operating System Fundamentals — NPTEL, IIT Kharagpur

9. ACADEMIC & COMPUTER SCIENCE FOUNDATION

Data Structures & Algorithms: Arrays, dynamic arrays, sorting, AVL trees, heaps, Horspool's algorithm, amortized complexity

Operating Systems: Deadlocks, semaphores, CPU scheduling, paging, TLB, Amdahl's law

Databases: SQL, relational databases, PostgreSQL, MySQL, MongoDB/BSON, indexing

Machine Learning Foundations: Regression, classification, clustering, PCA, SVD, model evaluation

Software Engineering: Agile, SDLC, Function Point Analysis, ROI, risk management, project management

Big Data: Hadoop, Spark, distributed processing and analytics

NLP / AI: TF-IDF, Word2Vec, embeddings, sentiment analysis, RAG, LangChain

10. PORTFOLIO SECTIONS TO BUILD

Section	What to show
Home	Headline, short introduction, primary CTA, GitHub/LinkedIn
About	Education, interests, career direction, strengths and development journey
Skills	Interactive skill groups covering Data, ML, AI, GenAI, Big Data, Backend and Tools
Projects	Featured projects with problem, solution, stack, screenshots/demo, GitHub and results
Experience	Internships, simulations, workshops or relevant practical experience
Certifications	Certificates with issuer, title, date and verification link where available
Education	Degree, institution, CGPA, expected graduation
AI / RAG	Dedicated section for RAG, LangChain, embeddings, vector search and LLM work
Contact	Email, LinkedIn, GitHub and a professional contact form

11. PORTFOLIO DESIGN DIRECTION

- **Visual identity:** modern developer/AI portfolio with clean typography, strong project cards and subtle technical visuals.
- **Hero:** lead with Data Science + AI/ML + Generative AI rather than a generic student introduction.
- **Projects:** make the RAG/LangChain project and Face Emotion Recognition project visually prominent.
- **Proof:** every major project should link to GitHub and, where possible, a live demo, screenshots or short technical case study.
- **Technical storytelling:** show the problem, architecture/workflow, technologies, result and lessons learned.
- **Recruiter flow:** Home → About → Skills → Projects → Certifications → Resume → Contact.

12. CONTENT STILL NEEDED FOR A TRULY COMPLETE PORTFOLIO

- Internship experience and company/project details, if any.
- Hackathons, coding competitions, workshops and technical events.
- Awards, academic achievements or leadership/positions of responsibility.
- LeetCode, HackerRank, CodeChef or other coding profiles.
- Exact GitHub repository links for each project.
- Live deployment links and project screenshots.
- Exact certificate dates, scores/grades and credential links.
- Additional projects that you want featured.
- A professional headshot, if you want a profile-photo section.

- Any publications, papers, blogs or technical articles.

13. PORTFOLIO POSITIONING

Primary positioning: Computer Science Engineer focused on Data Science, AI/ML and Generative AI.

Secondary positioning: Python / Backend Developer with strong data and AI capabilities.

Core differentiator: Combination of traditional data science and ML with modern RAG/LangChain/LLM workflows.

Recommended flagship story: Build a portfolio that demonstrates the progression from data analysis → machine learning → deep learning → NLP → RAG/Generative AI → deployable applications.

Source note: This profile incorporates the details from the uploaded resume and the additional RAG/LangChain and technical information you provided in our conversation. Items marked as needing more information should be verified before publishing.